

Kartikeya Krishna Mishra

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github.com/KartikeyaM2007 | LeetCode: 1690

PROFESSIONAL SUMMARY

AI/ML and Full Stack developer experienced in building **web applications**, **REST APIs**, and applied ML systems using **Next.js**, **PyTorch**, **FastAPI**, **Django**, **TensorFlow.js**, and **Node.js**. Skilled in **RAG**, **federated learning**, **computer vision**, **adversarial ML**, **IBM ART**, **agentic AI**, **hybrid retrieval**, **LLM reranking**, **secure authentication**, **RBAC**, and model evaluation.

EXPERIENCE

Defence Research and Development Organisation (DRDO)

India (Hybrid)

AI/ML Intern | IBM ART Security Testing

Jan. 2026 – April. 2026

- Performed **adversarial testing** on deep learning models using **IBM Adversarial Robustness Toolbox (ART)** to identify **model vulnerabilities**, perturbation behavior, robustness gaps, and security risks.
- Executed **adversarial attack and defense strategies**, improving model testing coverage by **40%**; documented findings for **AI reliability**, model evaluation, and critical-system validation.

NexCraft

India (Remote)

Software Development Intern

Jan. 2025 – Mar. 2025

- Developed and tested **5+ REST APIs** using **Node.js** for **CRUD operations**, JSON data exchange, backend workflows, API integrations, and SQL-backed services.
- Debugged **API integrations and SQL queries**; performed **Postman API testing**, input validation, edge-case testing, structured error handling, and reliability improvements.

PROJECTS

Anomaly Detection Vision Transformer | *PyTorch, VideoMAE, OpenCV, PySide6*

[App](#) | [Live](#) | [GitHub](#)

- Built** and trained a **9-Million parameter VideoMAE + AnomalyTransformer** on **250GB+ video data** from **Scratch**; extracted **768D visual features**, achieving **97.2% accuracy**, **0.98 AUROC**, and **<1s inference**.
- Packaged a Windows desktop app with **uploaded video analysis**, **live camera detection**, **batch prediction**, **anomaly timelines**, **score trends**, **detected frame grids**, and **event summaries**.

Federated Hospital System | *TensorFlow.js, Django, Federated Learning, RBAC*

[Live](#) | [GitHub](#)

- Developed a **privacy preserving federated learning** platform for hospitals to train models without sharing patient data; trained **5 models** on **140K+ samples**, reaching **92.69% accuracy**.
- Implemented **TensorFlow.js browser training**, Django aggregation APIs, RBAC dashboards, CSV validation, and centralized model aggregation with local data isolation.

GetFlowDone AI Agent Builder | *Next.js, React Flow, Convex, Gemini, Clerk*

[Live](#) | [GitHub](#)

- Created a visual agent workflow canvas with REST nodes, conditional logic, loops, tool chaining, parameter extraction, schema validation, and WebSocket autosave.
- Integrated **Gemini Flash**, **Convex DB**, **Clerk**, **Arcjet**, and Tailwind CSS; verified production build across **11/11 pages** with workflow execution and agent playground support.

RAG Based SHL Assessment Recommendation Engine | *FastAPI, Gemini, TF-IDF, RAG*

[Live](#) | [GitHub](#)

- Built a **hybrid retrieval pipeline** using FastAPI Async, Gemini embeddings, TF-IDF, NumPy vectors, metadata boosts, semantic search, and keyword cosine search.
- Added query expansion and **LLM reranking**, improving **Recall@10** from **0.373** to **0.65+** with lightweight **~2MB embeddings** and API key rotation.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, Tailwind CSS, React Flow

Backend: Node.js, Express.js, FastAPI, Django, Flask, REST APIs, JWT, RBAC

Full Stack: MERN Stack, Firebase, Convex DB

AI / ML: PyTorch, TensorFlow.js, Scikit-learn, Computer Vision, RAG, Federated Learning, Adversarial ML, Transformers, LLM Reranking

Tools: OpenCV, NumPy, Git, GitHub, Postman, Vercel, Railway, Hugging Face

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

HONOURS & ACHIEVEMENTS

LeetCode – Solved **250+ DSA problems**; achieved peak rating of **1690**.

Google Developer Student Club – Technical Core Member at Bennett University; contributed to technical activities and student developer initiatives.

Competitions – Secured **1st Runner-Up** in Ideathon and **3rd Rank** in Tech Arena Competition.

EDUCATION

Bennett University

Greater Noida, India

Bachelor of Technology in Computer Science and Engineering

Nov. 2023 – Jun. 2027

- Coursework:** Software Development, Machine Learning, Artificial Intelligence, Operating Systems, Computer Networks, DBMS, OOP, Data Structures & Algorithms

ADDITIONAL PROJECTS

Crux – Local Synthetic Data Engine | *FastAPI, PySide6, CTGAN, Ollama, Pandas*

[App](#) | [Web](#) | [GitHub](#)

- Built a **local-first CSV cleaning, repair, and synthetic data generation platform** with separate **web and desktop app versions**; integrated **FastAPI, PySide6, CTGAN, Pandas**, and local **Ollama/Llama 3** workflows.
- Implemented **query-aware generation chatbot, before-vs-after panels, PowerShell-style runtime logs, workflow visualizations, checksum-verified installer**, and CSV export pipelines for financial/fraud-style datasets.

LeLo.Interview – AI Interview Platform | *Next.js, TypeScript, Firebase, NextAuth*

[GitHub](#) | [Live](#)

- Developed a full-stack **AI-powered interview platform** with **resume-based interview generation, ATS scoring, AI feedback analysis**, and candidate evaluation dashboards.
- Integrated **Vapi AI voice agents**, secure authentication, protected routes, Firebase workflows, and responsive UI using **Tailwind CSS**.

HiringSpree AI – Agentic Resume Screening System | *LangGraph, Gemini, FAISS, BM25*

[GitHub](#) | [Live](#)

- Engineered an **agentic AI recruitment system** using **LangGraph, Gemini API, FAISS, BM25, hybrid retrieval**, and Reciprocal Rank Fusion.
- Built a **5-dimensional candidate evaluation engine** for skills, projects, education, experience, and communication fit with **Human-in-the-Loop review**, audit logging, and PII masking.

Fin-Mon – Expense Tracker | *MERN Stack, JWT, REST APIs, Recharts*

[GitHub](#)

- Built a responsive **MERN-based expense tracker** with **JWT authentication**, reusable UI components, validated backend workflows, and structured error handling.
- Integrated and tested **10+ REST APIs**; designed analytics dashboards using **Recharts**, reducing load time by **30%** across **5+ expense categories**.

ForgeRagAI – Custom RAG Pipeline | *Python, RAG, Hybrid Retrieval*

[GitHub](#)

- Developed a domain-adaptable **Retrieval-Augmented Generation** system for intelligent document search, Q&A, and retrieval-based generation workflows.
- Built modular retrieval logic for custom document querying, context selection, and response generation across domain-specific data.

Adversarial Attacks – ML Robustness Research | *Python, Jupyter, Adversarial ML*

[GitHub](#)

- Implemented adversarial attack experiments targeting machine learning models and analyzed model robustness against crafted perturbations.
- Evaluated model behavior under adversarial inputs using Python-based experimentation, visualization, and notebook-driven analysis.

Task Manager API | *Node.js, JavaScript, REST APIs*

[GitHub](#)

- Created a backend **REST API** for task management with CRUD operations, user authentication, and structured API workflows.
- Designed reusable routes and backend logic for task creation, update, deletion, and user-specific task handling.

Flask CRUD API Backend | *Python, Flask, REST APIs*

[GitHub](#)

- Built a Python **Flask REST API** with endpoints for create, read, update, and delete operations.
- Implemented backend request handling, response formatting, and API structure for lightweight CRUD-based services.